feb. 26

- Study guide posted, exam 1 week from today (3/5)
  - Covering Ch. 1–5 and start of Ch. 6 (last study guide bullet)
  - HW 5 still due 3/5

- To prove: the parity (even or odd) of a permutation's decomposition into 2-cycles (transpositions) is unique.

Ex $\alpha = (12357)(46)$
$= (17)(15)(13)(12)(46)$ 5 elements, so α is an odd permutation

- Lemma The identity is even i.e I can only write id as an even # of 2-cycles

Ex $id = (12)(12) = (12)(12)(13)(13) = (12)(13)(13)(12) =$ etc, but always even

Proof we proceed by induction. Case $n=0$ where $P(n)$ is "The identity cannot be written as a product of $2n+1$ 2-cycles"

for the base case, note $id \neq (ab)$.

Suppose the id cannot be written as a product of $2n+1$ 2-cycles for some $n \geq 0$.

By way of contradiction, suppose $id = \tau_1 \tau_2 \cdots \tau_{2n+1} \tau_{2(n+1)} \tau_{2(n+1)+1}$ where the τ_i are 2-cycles.

focus on $\tau_{2(n+1)} \tau_{2(n+1)+1}$. The four possibilities are

$\tau_{2(n+1)} \tau_{2(n+1)+1} = (ab)(ab)$ case (1)
$= (cd)(ab)$ case (2)
$= (ab)(ac)$ case (3)
$= (bc)(ab)$ case (4)

We are interested in moving a to the left.

(1) If this is the case $\tau_{2(n+1)} \tau_{2(n+1)+1} = id$, so $id = \tau_1 \cdots \tau_{2n+1}$ contradicts hypothesis

(2) $= (ab)(cd)$
(3) $= (acb) = (ac)(cb)$
(4) $= (acb) = (ac)(cb)$

We repeat argument with $\tau_{2n+1} \tau_{2(n+1)}$. We either get a contradiction or we move a to the left.

Continuing, we either get a contradiction or τ_1 is the only 2-cycle with a. In the latter case, we have a contradiction since $a = id(a) \neq \tau_1(a)$. We finish by invoking the axiom of induction. $\square$

- Thm. The parity (even or odd) of a permutation's decomposition into 2-cycles (transpositions) is unique.

Proof

By way of contradiction, suppose $\beta = \tau_1 \ldots \tau_t = \sigma_1 \cdots \sigma_s$ where the τ_i and σ_i are transpositions, t is odd and s is even. Then $\tau_1 \cdots \tau_t = \sigma_1 \cdots \sigma_s$, so $id = \tau_1 \cdots \tau_t \sigma_s \sigma_{s-1} \cdots \sigma_1$

$= \underset{\text{odd}}{\text{s.t.}}$ 2-cycles, which is a contradiction. $\square$

- Let $A_n = \{\sigma \in S_n : \sigma \text{ is even}\}$ This is the alternating group

This is Ch.5 exercise 17 on HW 4

- **Thm.** $A_n < S_n$ and $|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}$, $n > 1$

  Proof

  Note $\overset{n!}{|S_n|} < \infty$ and $id \in A_n$ by the previous lemma, so we apply the finite subgroup test. We need only show closure. Let $\sigma, \tau \in A_n$, so $\sigma = t_1 \cdots t_r$ and $\tau = T_1 \cdots T_v$ where r, v are even. We see $\sigma\tau = t_1 \cdots t_r T_1 \cdots T_v$. Since $r+v$ is even, we see $\sigma\tau \in A_n$. We are done by finite subgroup test.

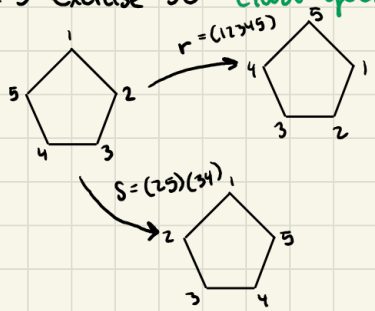
  $A_n \hookrightarrow \{\text{odd permutations}\}$

  $\sigma \longmapsto \sigma(12)$

  this is our bijection

  So $|A_n| = |\{\text{odd permutations}\}|$ which together make up S_n, so each is half of $|S_n|$. $\square$

- Ch.5 exercise 50 (last question on HW 4)



Notice D_n and S_n are isomorphic